

BALL TO BALL DATA ANALYSIS

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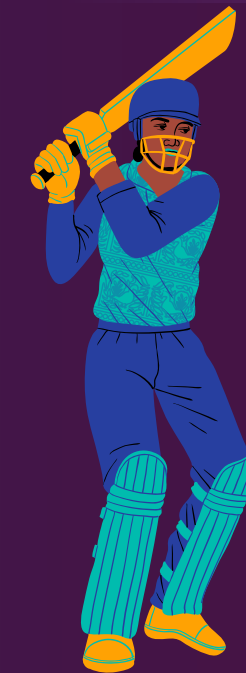
ABSTRACT



Collected a structured cricket match dataset from Kaggle. This dataset provided historical match data which served as the foundation for performing data cleaning, analysis, and visualization using various data analytics tools and techniques.



Cleaned, transformed, and analyzed the scraped data using Pandas to ensure structured formatting, accuracy, and readiness for visualization and reporting. Utilized Pandas to clean, structure, and analyze the scraped data, ensuring it was accurate, consistent, and ready for visualization.



Built interactive Power BI dashboards to visualize player stats, team performance, and match insights, enabling better interpretation and decision-making through data storytelling. Designed interactive Power BI dashboards to visualize key cricket metrics, including player stats, team comparisons, and match insights.

PROBLEM STATEMENT

BUILD THE UNBEATABLE XI — WHERE EVERY PLAYER IS A GAME-CHANGER!

**1. THE TEAM SHOULD BE ABLE TO SCORE ATLEAST
180 RUNS ON AVG.**

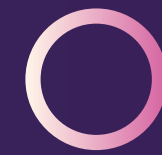
**2. THEY SHOULD BE ABLE TO DEFEND 150 RUNS ON
AVERAGE.**



OBJECTIVE



To perform end-to-end data analysis on T20 World Cup 2024 player statistics.



To select an ideal playing XI based on key performance metrics.



To visualize performance insights using Power BI dashboards.



To work without web scraping by using pre-existing datasets or manual data collection.

PROPOSED SYSTEM



COLLECT T20 WORLD CUP 2022 DATA FROM PUBLIC DATASETS.



CLEAN AND PROCESS THE DATA USING PYTHON AND PANDAS.



ANALYZE KEY PERFORMANCE INDICATORS LIKE RUNS, STRIKE RATE, ECONOMY, WICKETS, ETC.



USE POWER BI TO CREATE DASHBOARDS FOR VISUAL ANALYSIS.



SELECT THE BEST XI PLAYERS BASED ON BALANCED PERFORMANCE CRITERIA.

ROLES AND SELECTION CRITERIA

OPENERS



PARAMETERS	DESCRIPTION	CRITERIA
Batting Average	Average runs scored in an innings	> 30
Strike Rate	No of runs scored per 100 balls	> 140
Innings Batted	Total Innings batted	> 3
Boundary %	% of runs scored in boundaries	> 50
Batting Position	Order in which the batter played	< 4

ROLES AND SELECTION CRITERIA

ANCHORS / MIDDLE ORDER



PARAMETERS	DESCRIPTION	CRITERIA
Batting Average	Average runs scored in an innings	> 40
Strike Rate	No of runs scored per 100 balls	> 125
Innings Batted	Total Innings batted	> 3
Avg. Balls Faced	Average balls faced by the batter in an innings	> 20
Batting Position	Order in which the batter played	> 2

ROLES AND SELECTION CRITERIA

FINISHER / LOWER ORDER ANCHOR



PARAMETERS	DESCRIPTION	CRITERIA
Batting Average	Average runs scored in an innings	> 25
Strike Rate	No of runs scored per 100 balls	> 130
Innings Batted	Total Innings batted	> 3
Avg. Balls Faced	Average balls faced by the batter in an innings	> 12
Batting Position	Order in which the batter played	> 4
Innings Bowled	Total Innings Bowled by the bowler	> 1

ROLES AND SELECTION CRITERIA

ALL-ROUNDERS / LOWER ORDER



PARAMETERS	DESCRIPTION	CRITERIA
Batting Average	Average runs scored in an innings	> 15
Strike Rate	No of runs scored per 100 balls	> 140
Innings Batted	Total Innings batted	> 2
Batting Position	Order in which the batter played	> 4
Innings Bowled	Total Innings bowled	> 2
Bowling Economy	Average runs allowed per over	< 7
Bowling Strike Rate	Average no. of balls required to take a wicket	< 20

ROLES AND SELECTION

CRITERIA

SPECIALIST FAST BOWLERS

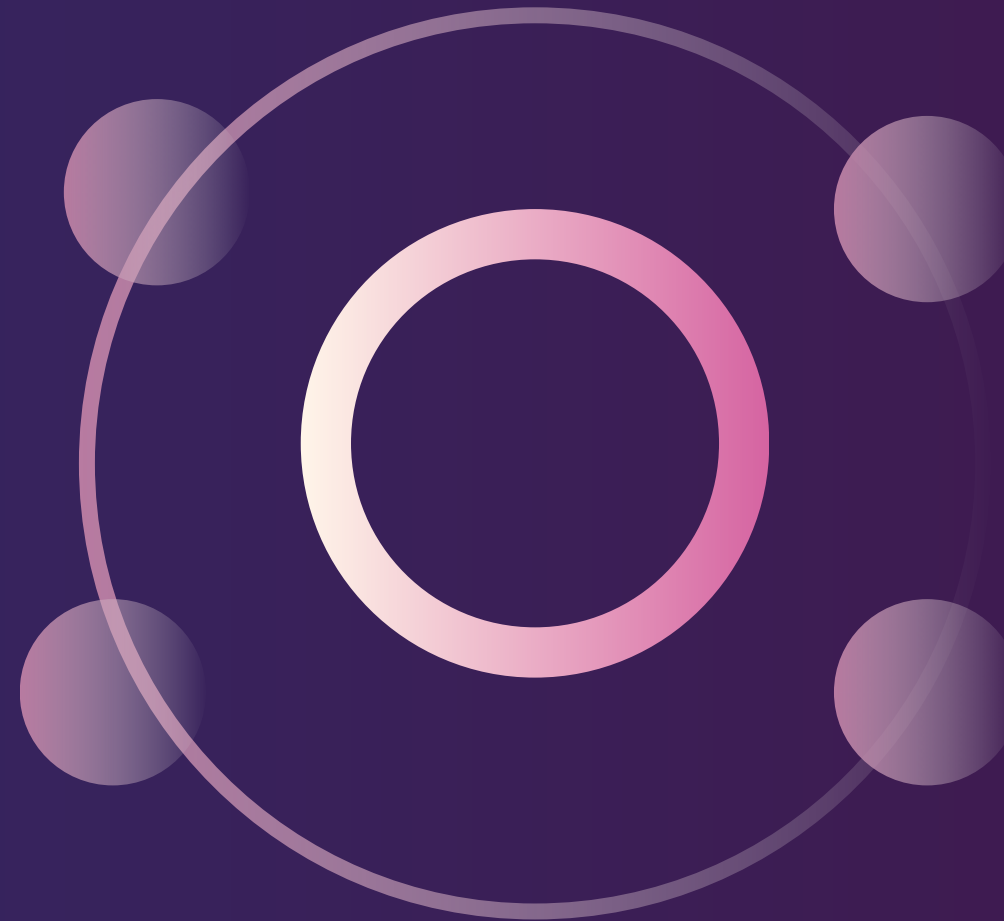


PARAMETERS	DESCRIPTION	CRITERIA
Innings Bowled	Total Innings bowled	> 4
Bowling Economy	Average runs allowed per over	< 7
Bowling Strike Rate	Average no. of balls required to take a wicket	< 16
Bowling Style	Bowling style of the player	= "%Fast%"
Bowling Average	No. of runs allowed per wicket	< 20
Dot Ball %	% of dot balls bowled	> 40

ADVANTAGES

Insightful Analysis: Delivers meaningful insights into player and team performance using historical data, helping users understand trends and key match dynamics.

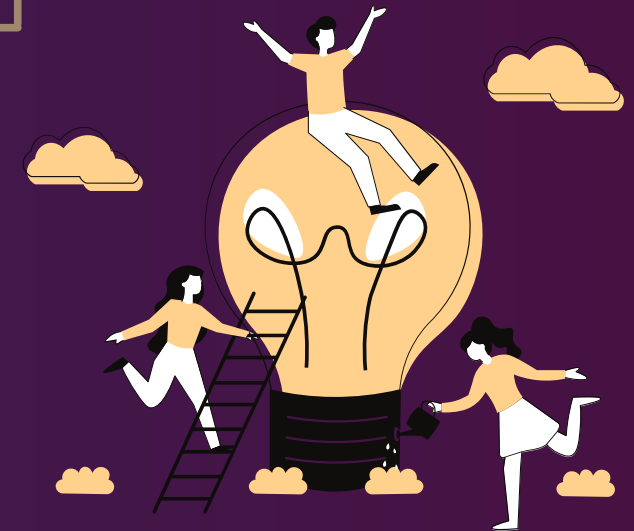
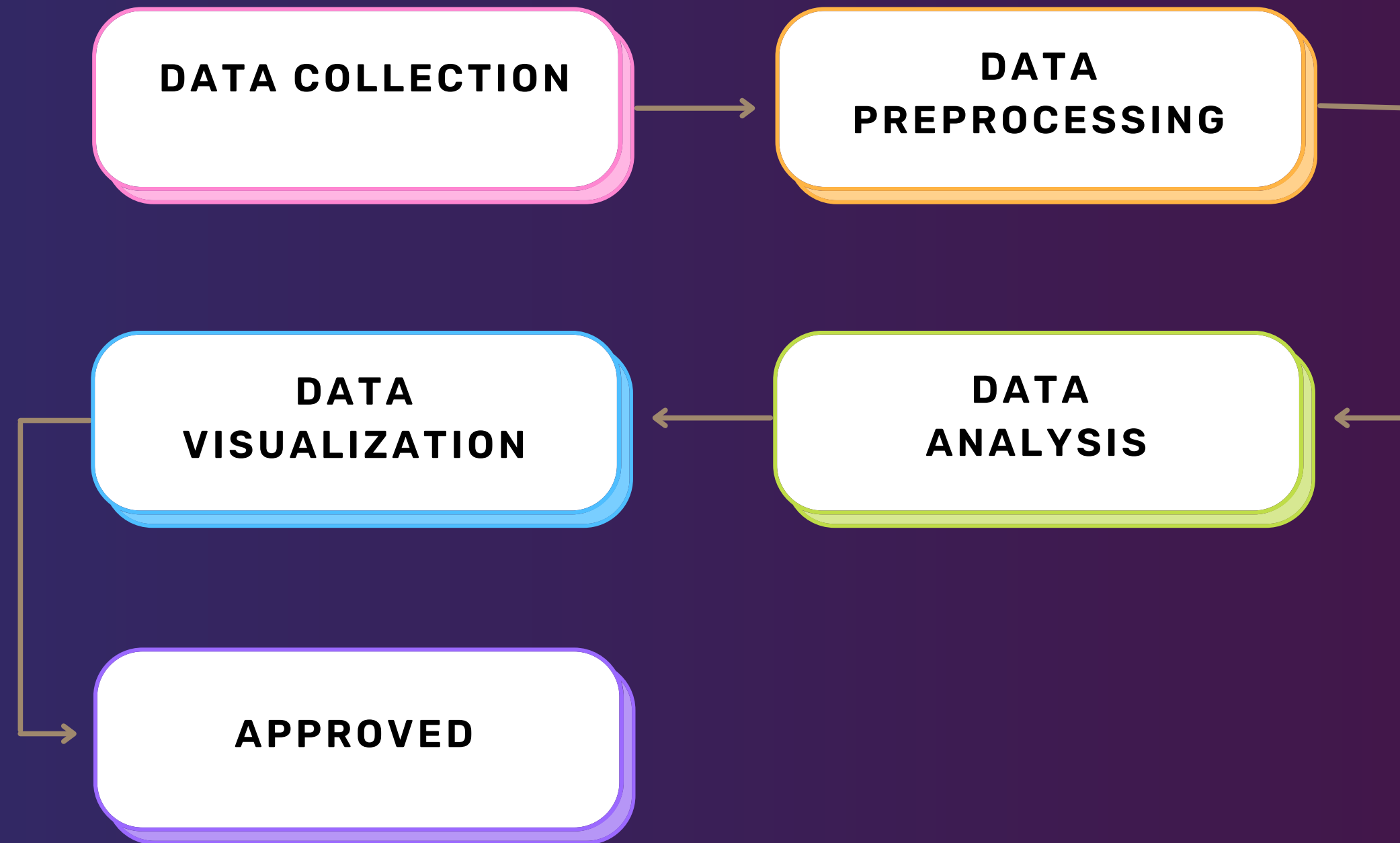
Interactive Dashboards: Power BI visualizations allow users to explore data through filters, comparisons, and real-time updates, enhancing user engagement.



Time-Efficient Workflow: Utilizes a well-structured Kaggle dataset, eliminating the need for complex data scraping and saving valuable development time.

Practical Skill Application: Demonstrates real-world use of Python, Pandas, and Power BI—essential tools for modern data analytics and visualization projects.

SYSTEM ARCHITECTURE



MODULES



DATA COLLECTION MODULE

Cricket data from Kaggle, including match details, player stats, and outcomes



DATA PREPROCESSING MODULE

Cleans, formats, and structures the raw dataset using Python and Pandas



DATA ANALYSIS MODULE

statistical computations to extract KPIs, player performance metrics, win rates, and match trends.



VISUALIZATION MODULE

Uses Power BI to create interactive dashboards for data exploration, filtering, and insight generation.



SOFTWARE SPECIFICATIONS

**Programming
Language**

Python

**Visualization
Tool**

Power BI

**Dataset
Source**

Kaggle

**Libraries
Used**

Pandas

IDE

Google
Colab



RUN TIME? MORE LIKE PANIC TIME

"Featuring: Unexpected errors and last-minute breakthroughs."



<https://colab.research.google.com/drive/16AoOEh1bsojpH-Lo1RyLhtSiWbMW/mlFa?usp=sharing>



THANK YOU!

DATA ANALYSIS IS KEY TO BUSINESS
GROWTH AND SUCCESS!